

SMART SERVICE & REPAIR MANAGEMENT SYSTEM

10 DAY FULL BACKEND + FRONTEND INTEGRATION ROADMAP

PROJECT GOAL

Build a real industry level Smart Service & Repair Management System covering:

- Authentication (Register/Login)
- Role Based Access (Admin, Manager, Technician, Customer)
- Customer Management
- Service Ticket Management
- File Upload (Image + PDF)
- MongoDB Advanced Operators
- Inventory Management
- Invoice PDF Generation
- Email Sending

INDUSTRY LEVEL BACKEND FOLDER STRUCTURE

backend

src

config

db.js

models

User.js

Customer.js

Service.js

Part.js

Invoice.js

controllers

authController.js
customerController.js
serviceController.js
partController.js
invoiceController.js
routes
authRoutes.js
customerRoutes.js
serviceRoutes.js
partRoutes.js
invoiceRoutes.js
middlewares
authMiddleware.js
roleMiddleware.js
uploadMiddleware.js
utils
generateToken.js
sendMail.js
pdfGenerator.js
app.js
server.js

DAY 1 – REGISTER MODULE

Backend Tasks:

- User Schema Create
- Register API
- Bcrypt Password Hash
- Image + PDF Upload

Endpoint:

POST /api/auth/register

Postman JSON Body:

```
{  
  "name": "Prateek",  
  "email": "prateek@gmail.com",  
  "password": "123456",  
  "role": "ADMIN",  
  "phone": "9876543210"  
}
```

Frontend:

- Register Form
- File Upload
- Axios Integration

DAY 2 – LOGIN + JWT + COOKIE

Backend:

POST /api/auth/login

Postman:

```
{  
  "email": "prateek@gmail.com",  
  "password": "123456"  
}
```

Frontend:

- Login Page
- Protected Routes
- Role Based Dashboard

DAY 3 – CUSTOMER MODULE

Schema Fields:

name, phone, address, deviceName, devicePrice, purchaseBill(PDF), deviceImage(Image)

Endpoints:

POST /api/customers

GET /api/customers

GET /api/customers/:id

Frontend:

- Customer Form
- Customer Table

DAY 4 – SERVICE TICKET MODULE

Schema:

customerId, problemDescription, serviceCharge, status, images[], reports[]

Endpoints:

POST /api/services

GET /api/services

PUT /api/services/:id

Frontend:

- Create Ticket
- Service Table

DAY 5 – MONGODB UPDATE OPERATORS

Use:

\$set

\$push

\$pull

\$unset

\$addToSet

Endpoint:

PUT /api/services/update-status/:id

DAY 6 – TECHNICIAN ASSIGNMENT

Endpoint:

PUT /api/services/assign-tech/:id

DAY 7 – FILE UPLOAD ADVANCED

Upload:

- Service Proof Image
- Repair Report PDF

Endpoint:

POST /api/services/upload-doc/:id

DAY 8 – INVENTORY + BULK UPLOAD

Endpoints:

POST /api/parts

POST /api/parts/bulk-upload

Sample Bulk JSON:

```
[  
  {"partName":"Screen","stock":20,"price":5000},  
  {"partName":"Battery","stock":15,"price":3000}  
]
```

DAY 9 – INVOICE PDF GENERATION

Endpoint:

POST /api/invoice/generate/:serviceId

Frontend:

- Generate PDF Button
- Download PDF

DAY 10 – EMAIL SENDING

Endpoint:

POST /api/invoice/send-mail/:invoiceId

Frontend:

- Send Email Button

SCHEMA DATA TYPES COVERED

String

Number

Image File

PDF File

Array

Nested Object

LEARNING OUTCOME

Real Backend Architecture

MongoDB CRUD + Operators Mastery

Authentication + RBAC

File Handling

PDF Generation

Email Integration

Industry Ready Project Experience
